
DARIO MARINO

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EDUCATION

University of Chicago

Master's in Computational Social Science-Economics (MACSS-Econ), September 2024 - June 2026

- GPA: 3.61
- Thesis: *Of Rivalry and Synergy: Nonlinear Patents and Products Similarity Effects on R&D Investment*
Supervisor: Ufuk Akcigit

University of Bologna Alma Mater Studiorum

2 Graduate and 4 Undergraduate Non-Degree Math Coursework September 2023 - July 2024

- Graph Theory; Stochastic Processes; Linear Algebra (from Bachelor's Degree in Electrical Engineering)
- Real Analysis 1A; Real Analysis 1B; Probability and Mathematical Statistics

Paris School of Economics (PSE - Ecole d'économie de Paris)

Master's Degree in Analysis and Policy in Economics, September 2022 - June 2023

- Final Average: 13.6/20 [GPA: 3.4 ; B+]; Thesis Grade: 14/20; [With Honors]
- Thesis: *Balanced Innovation: Nonlinear optimal patent breadth in an interaction of demand and technology networks*; Supervisor: Francis Bloch

University of Paris (Université Paris 1), Panthéon Sorbonne

Panthéon Sorbonne Master of Economics, September 2021 - June 2022

- Average Grade: 16/20 [GPA: 3.7 ; A-]; Thesis Grade: 18/20
- Thesis: *The Overlooked Effect of Prices on Productivity and Pay Gap*; Supervisor: Carmen Camacho-Perez

University of Bologna (Università di Bologna) Alma Mater Studiorum

Bachelor's Degree in Political Science and International Relations, L-36, September 2018 – December 2021

- Average Grade: 110/110 with Honors. [GPA: 4.0 ; A+];
- Thesis: *The Effects of Automation on Labor Market*; Supervisor: Antonello Eugenio Scorcu

RESEARCH EXPERIENCE

Knowledge Lab at UChicago

Supervisor: Austin Kozlowski

Research Assistant, June 2025 - Present

- Worked in collaboration with the ART Lab (Akcigit Research Team) on datasets of product similarity and patent data
- Developed Semiconductor Dataset with product description embeddings, manufacturer-assignee matching for USPTO patents, and extraction of product characteristics from specs documents. Applied NLP techniques and ChatGPT API.
- Fully designed a pipeline that uses Open Router prompts to connect with several AI models (Deepseek, Kiwi, Mistral, Google, and OpenAI), which score patents based on their relevance to product descriptions.
- Built a teacher-student distillation local deep learning model to mimic LLM responses for scoring product-patent pairs on usage, technical importance, and relevance. The system, which used a hybrid feature set and XGBoost regressors, achieved significant cost reductions while maintaining strong alignment with the LLM's judgments.
- The Knowledge Lab is directed by James Evans and the project is funded by the NSF grant "Assessing and Predicting Technology Outcomes" (APTO). Full time position.

Research Assistant for Philippe Aghion and Antonin Bergeaud

Supervisor: Antonin Bergeaud

Research Assistant, June 2025 - September 2025

- Extended the measure developed in “Measuring Technological Innovation Over the Long Run” by Kelly, Papanikolaou, Seru, and Taddy (2021), using PatentSBERTa_V2 embeddings instead of tf-idf. This new patent novelty dataset will be used for another working paper by Philippe Aghion and Antonin Bergeaud.
- The measures compute abstracts and claims pairwise similarity for each patent with respect to all the other patents in the interval 5 years before and 5 years after. The patents that have larger similarity with the future patents rather than the past patents are the more novel ones.
- Parallelized similarity computations across a 128 core remote server to efficiently manage large-scale data and operations. Analyzed all EPO patent abstracts (1978-2025) and USPTO abstracts and claims (1836-2025).
- Position funded by Philippe Aghion’s Economics of Institutions, Innovation and Growth Chair at Collège de France.

Research Assistant for Gianmarco Ottaviano

Supervisor: Gianmarco Ottaviano

Research Assistant, June 2025 - September 2025

- Collaborated with professor Gianmarco Ottaviano from Bocconi on the co-authored project: “Worldwide Negative Externalities from Regulation: Selective Availability Repeal Effects on GPS Precision Farming”
- UN Comtrade dataset for imports, SPAM dataset for agricultural outputs. Constructed a Starlink coverage dataset
- Performed regressions on the effect of agricultural robot imports on agricultural yield, hectares and production for crops suited for precision farming.

Akcigit Research Team at UChicago (ART Lab)

Supervisor: Ufuk Akcigit

Research Assistant, December 2024 - June 2025

- Worked on patent description and patent claims similarity between incumbent and follower firms using a calibrated BERT model. Increase in patent text similarity between firms points to a reduction in creative destruction at the R&D level
- Analyzed how patent concentration affects entrants’ innovation by applying disruption scores (Lin, Evans, Wu 2021) computed from citation networks. The scores measure whether a patent advances prior art or remains in the previous network of science.
- Used NLP data on patent abstract to compute an innovation score for all the patents in USPTO dataset.
- Replicated and extended concentration measures in Akcigit & Ates (2023), including top 1%, top 3, and top 5 firm measures at different IPC aggregation levels.
- Estimated fixed-effects regressions on lagged concentration and entrants’ disruption scores, documenting significant negative effects of concentration on innovation.
- Datasets from USPTO. Research for Ufuk Akcigit and Daron Acemoglu

Structural Change and Global Dynamics (EU grant)

Supervisors: Alireza Jay Naghavi (University of Bologna); Antonio Minniti (University of Bologna)

Research Fellowship, June 2023 - September 2024

- Wrote and implemented all microeconometrics coding with R and webscraping with Python
- Conducted Ownership Networks and Environmental Analysis of European firms using Moody’s Orbis dataset
- Full time research position designed for post-docs in collaboration with Bocconi and the University of Turin.

A Generalized Measure of Bargaining Power with an Application to EU Law-Making (Bocconi University)

Supervisors: Joseph-Simon Görlach (Bocconi); Nicolas Motz (Universidad Complutense de Madrid)

Research Assistant, November 2022 - March 2023

- Web Scraping with Python of European parliament laws both on html and json to pdf
- Performed Sentiment Analysis with R on all the European parliament laws in the dataset

WORKING PAPERS/CHAPTERS

- **Dario Marino** and Nicola Rossi. “Economic Culture.” In *Un miracolo non fa il santo, la cultura degli italiani*, edited by Nicola Rossi. Rome: IBL Libri, 2025.

PROJECTS/WORK IN PROGRESS

- **Dario Marino** and Gianmarco Ottaviano. *Worldwide Negative Externalities from Regulation: Selective Availability Repeal Effects on GPS Precision Farming*

TEACHING EXPERIENCE

University of Chicago

Teaching Assistant | September 2024-January 2025

- Economic Analysis of Law; Undergraduate; Public Policy Studies; Jim Leitzel

University of Bologna Alma Mater Studiorum

Teaching Assistant | August 2023-July 2024

- Monetary Policy; Bachelor's Degree in Economics, Markets and Institutions; Antonello Eugenio Scorcu
- Public Economics; Master's Degree in Economics and Public Policy; Daniele Fabbri
- Statistics; Master's Degree in Law, Economics and Governance; Luke Brian Connelly
- Econometric Methods for Policy Evaluation; Master's Degree in Economics and Public Policy; Guglielmo Barone
- Economics of Competition; Master's Degree in Law and Economics; Emanuela Carbonara
- Statistics and Programming; Bachelor's Degree in Economics, Politics and Social Sciences; Laura Anderlucci

ADDITIONAL EXPERIENCE

“Liberi oltre le illusioni” (Think Tank & YouTube Series) ||| <https://www.youtube.com/watch?v=JZJMR4AJQ-U>

Content Creator, March 2022-Present

- Produced a YouTube series on economic growth and innovation with Prof. Michele Boldrin (Washington University in St. Louis); curated two series on economic literature pills covering innovation and the minimum wage
- Free market oriented think tank with more than 80,000 subscribers on Youtube.

“The Optimizers” – Econ Substack Newsletter ||| <https://theoptimizers.substack.com/>

Content Creator, September 2023-Present

- Write short applied economics pieces on innovation, growth, and productivity

HONORS & AWARDS

Maroon Research Scholarship University of Chicago: August 2024 - July 2026

Mercatus Center Frédéric Bastiat Fellowship June 2025 - May 2026

Mercatus Center Don Lavoie Fellowship: August 2024 - June 2025

LANGUAGE SKILLS & CERTIFICATIONS

English: TOEFL iBT, Nov 2025 - 105 - 26 R/ 29 L/ 23 S / 27 W | Aug 2023 - 111 - 28 R/ 29 L/ 27 S/ 27 W

French: Course (4 ECTS, University of Bologna) 2021; Course (4 ECTS, Université Paris 1 Panthéon Sorbonne) 2022

GRE: Quantitative: 160; Verbal: 159 - September 2022

PROGRAMMING

- Python [IDE: Spyder] (Specialized)
- R [IDE: RStudio] (Specialized)
- STATA (Specialized)
- MATLAB (Specialized)
- Mathematica (Intermediate)

REFERENCES

Ufuk Akcigit

Title: Master Thesis Supervisor and Research Supervisor

University of Chicago; uakcigit@uchicago.edu

Gianmarco Ottaviano

Title: Research Supervisor and Project Co-Author;

Bocconi University; gianmarco.ottaviano@unibocconi.it

Antonin Bergeaud

Title: Research Supervisor

Centre for Economic Performance at the LSE, Innovation Lab at Collège de France, HEC Paris; bergeaud@hec.fr